

torch.linalg.det#

<https://pytorch.org/docs/stable/generated/torch.linalg.det.html>

Description:

Computes the determinant of a square matrix. Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions. `torch.linalg.slogdet()` computes the sign and natural logarithm of the absolute value of the determinant of square matrices. A (Tensor) – tensor of shape $(*, n, n)$ where $*$ is zero or more batch dimensions.

Function Signature

`torch.linalg.det(A, *, out=None) → Tensor#`

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – output tensor. Ignored if None. Default: None.

Code Examples

```
>>> A = torch.randn(3, 3)
>>> torch.linalg.det(A)
tensor(0.0934)

>>> A = torch.randn(3, 2, 2)
>>> torch.linalg.det(A)
tensor([1.1990, 0.4099, 0.7386])
```

Notes & Warnings

NOTE See also `torch.linalg.slogdet()` computes the sign and natural logarithm of the absolute value of the determinant of square matrices.

Detailed Documentation

Documentation

Computes the determinant of a square matrix.

Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions.

`torch.linalg.slogdet()` computes the sign and natural logarithm of the absolute value of the determinant of square matrices.

A (Tensor) – tensor of shape $(*, n, n)$ where $*$ is zero or more batch dimensions.

out (Tensor, optional) – output tensor. Ignored if None. Default: None.